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KOSUKE UKITA, Kyushu Institute of Technology, Kitakyushu, Fukuoka, Japan

TSUYOSHI OKITA, Kyushu Institute of Technology, Kitakyushu, Fukuoka, Japan

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Analysis of Human Activity Recognition by Diffusion Models

Kosuke Ukita
Kyushu Institute of Technology
Department of Artificial Intelligence
Iizuka, Fukuoka, Japan
ukita.kosuke299@mail.kyutech.jp

Tsuyoshi Okita
Kyushu Institute of Technology
Department of Artificial Intelligence
Iizuka, Fukuoka, Japan
tsuyoshi@ai.kyutech.ac.jp

Abstract

This paper proposes a classifier based on a diffusion model for human activity recognition. To this end, we introduce three architectures: (1) the representation-conditional diffusion transformer, (2) the first classifier, RepcondFormer, and (3) the second classifier, RepcondClassifier. Experimental results show that RepcondFormer outperforms in three datasets, while RepcondClassifier performs better in one dataset when fine-tuned.

CCS Concepts

• **Computer systems organization → Machine learning algorithms**.

Keywords

Generative Model, Diffusion Model, Human Activity Recognition

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1 Introduction

Recent advances in large-scale text-to-image diffusion models, such as DALL-E 2 [11], have dramatically demonstrated the capabilities of diffusion models in generative AI. More recently, the diffusion transformer (DiT) [10] has further enhanced the quality of the generative process by replacing the UNet module with a transformer module.

Building on the generative processes learned using diffusion models, a natural progression is to apply these generative processes to classification tasks. Han et al. [2] proposed a classification and regression diffusion model that integrates a classifier within the diffusion model itself. In this model, the classifier is trained concurrently with the diffusion processes. Li et al. [7] introduced a method for extracting a zero-shot classifier from a class-conditional diffusion process learned by the diffusion models. Their approach involves a two-step process: first, learning the generative process, and second, constructing a classifier based on this generative process. Thus, once high-performance diffusion models are obtained, focus can shift to optimizing the classifier. Ukita et al. [5] presented the concept of representation-conditional diffusion models, which outperform class-conditional diffusion models. This approach has been applied to medical CT images.

Research on generative models for one-dimensional signals, such as Human Activity Recognition (HAR), remains limited. First, a major category of models includes language transformers, like Large Language Models (LLMs) such as GPT-4 [9]. These models are being applied in various contexts, such as dataset contamination detection [3]. Second, an extension of vision transformers has been applied to HAR tasks [8]. Third, diffusion models are utilized for signal generation. For instance, Zuo et al. [23] employ a diffusion model to generate one-dimensional signals for HAR, outperforming TimeGAN [21], which uses GAN technology. However, to our knowledge, no diffusion models have yet been applied to signal classification.

The contributions of this paper are as follows:

- We propose a representation-conditional diffusion transformer specifically designed for one-dimensional signals, such as accelerometer sensor input, rather than medical images.
- We introduce the classifier RepcondFormer, which is based on the transformer encoder of the diffusion transformer and is applied to HAR tasks.
- We present the classifier RepcondClassifier, which is equipped with a class-conditional diffusion model generated from the representation-conditional diffusion classifier.

This paper is organized as follows: Section 2 reviews related research, Section 3 presents our proposed model, Section 4 discusses the experimental results, and Section 5 concludes the paper.

2 Related Research

Vision Transformer Approaches. Okita et al. [8] proposed a vision transformer model for handling sensor data in HAR. This model employs a multitask self-supervised learning approach (or pre-trained language model) based on the vision transformer. The training objectives utilize a masked image model, with 70% masking proving to be the most effective for HAR. Additionally, the model introduces pretext tasks that involve seven types of masks: permutation, time warping, rotation, WDBA, RGW, and DGW. To the best of our knowledge, no other models have applied vision transformers to HAR.

Diffusion Transformer Approaches. The Latent Diffusion Model [14] is a pixel-level image model utilizing VAE [4]. DiT [10] is based on

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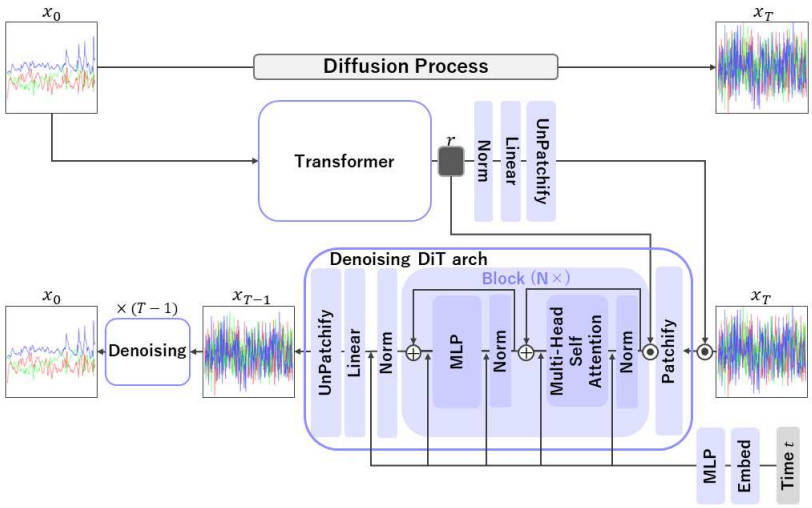

Figure 1: The figure illustrates a representation-conditional diffusion transformer. The upper part shows the diffusion process for injecting noise, depicting the mechanism for adding noise. The central part represents the conditioning mechanism using the representation. The representation r is generated by the Transformer's encoder. This representation r follows two paths. The first path is decoded through norm, linear, and unpatchify layers, reaching the x_T-th sliding window (refer to the right line from r). The second path preserves the dimensions of representation r until it reaches the line just before the Block in the denoising networks (refer to the downward line from r). The lower part displays the denoising network, constructed similarly to DiT but for one-dimensional signals rather than images. Note that the figure depicts only the transition from the x_T -th to the x_{T-1}-th sliding window. This process is applied repeatedly in all denoising steps from x_T to x_{T-1}.

the denoising networks of the Transformer [18]. Among DiT models, the largest variant, DiT-XL/2, achieves a state-of-the-art FID score of 2.27, outperforming other diffusion models.

3 Our Proposed Model

The proposed model consists of a representation-conditional diffusion transformer and two types of classifiers.

3.1 Representation-conditional Diffusion Transformer

Figure 1 illustrates our representation-conditional diffusion transformer for sensor input. Our model is based on the class-conditional diffusion transformer provided by DiT, which uses a Transformer architecture instead of UNet. The representation-conditional diffusion transformer integrates a transformer encoder that processes sensor data input, encoding a sliding window into the representation r. In Figure 1, r is depicted between the Transformer and Norm components.

DiT was trained with an input size of $32 \times 32 \times 4$, operating in the latent space where images are compressed. In contrast, our representation-conditional diffusion transformer was trained with an input size of 300×3 sliding windows, without utilizing the latent space.

After training our representation-conditional diffusion transformer, we explore two approaches in the subsequent experiments. The first approach involves building a classifier, named RepcondFormer, by fine-tuning the internal Transformer encoder, as detailed in the next subsection. The second approach involves constructing the classifier RepcondClassifier, which generates classification results from a diffusion transformer tuned with class-conditional labels, as discussed in the following subsections.

3.2 Classification with RepcondFormer

The representation-conditional diffusion transformer integrates a transformer that encodes the sliding window into a representation (see Figure 1). By connecting this representation to the denoising network, it is possible to generate signals conditionally. This representation is optimized to facilitate the training of the diffusion model, rather than simply adding a pre-trained fixed representation. The goal of training the diffusion model is to predict the source noise within the noisy sliding window, effectively separating noise from the signal. Using representation in place of class labels empirically simplifies the training process.

We adapted the class-conditional diffusion transformer into a representation-conditional diffusion transformer and developed a classifier based on this new model, which we have named RepcondFormer. It is important to note that RepcondFormer can be fine-tuned in two ways: transfer learning and full fine-tuning. In transfer learning, only the final layer of the classifier is updated while keeping the pre-trained parameters fixed. In contrast, full fine-tuning involves updating all parameters, starting with the initial values after pre-training.

3.3 Classification with RepcondClassifier

Since the diffusion model handles internal information as noise and does not directly represent signals, it is generally not considered an encoder that outputs representations. To address this, we

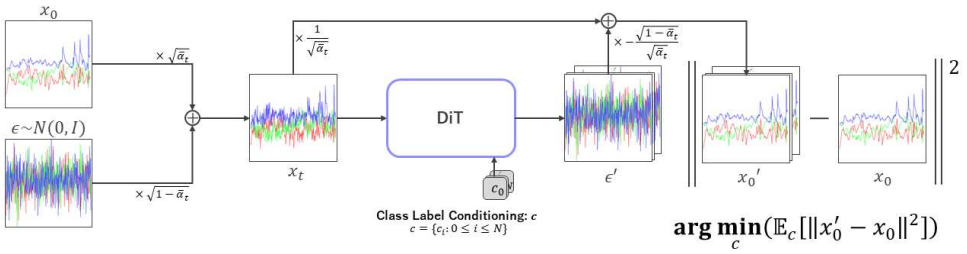

Figure 2: The figure provides an overview of RepcondClassifier. DiT is trained to predict the source noise ϵ present in the received signal x_T. During this process, all class labels c are conditioned, and ϵ', which estimates the source noise ϵ, is generated for each class. The predicted original signal x_0' is derived from x_T and ϵ', with the closest match to the original signal being the predicted classification result.

developed the RepcondClassifier, as shown in Figure 2, inspired by the *Diffusion Classifier* [7].

The Diffusion Classifier can be applied to class-conditional diffusion models. It separates the source noise ϵ and the signal x_0 from the noisy sliding window x_T according to the time step T. After predicting the signal x_0' using the class-conditional diffusion models, it calculates the average squared error between the predicted signal x_0' and the original signal x_0. By examining which signal class is most accurately restored, classification results can be obtained.

In the original *Diffusion Classifier*, ϵ' predicts the source noise ϵ from the noisy image x_T. The squared error is then measured between ϵ' and ϵ. In contrast, our RepcondClassifier does not use noise but instead calculates the squared error between the sliding window x_0' (predicted after removing the noise) and the original signal x_0.

This process involves the following three steps to utilize a class-conditional diffusion model:

(1) Train the representation-conditional diffusion transformer.
(2) Restore the trained denoising network conditioned on class labels.
(3) Obtain results using RepcondClassifier.

4 Experimental Results

In the first subsection, we describe the datasets used in these experiments. Next, we present the results for the two classifiers: RepcondFormer and RepcondClassifier. Finally, we discuss the evaluation of these classifiers and the signal generation processes.

4.1 Dataset

Table 1: Datasets we used in this experiments

Dataset	Subjects	Samples	Classes	References
Capture-24	152	573K	4	Willetts et al.[20]
ADL	7	0.6K	5	Bruno et al.[1]
Opportunity	4	3.9K	4	Roggen et al.[13]
PAMAP2	8	2.9K	8	Reiss and Stricker[12]
REALWORLD	14	12K	8	Sztyler and Stuckenschmidt[17]
WISDM	46	28K	18	Weiss et al.[19]

We used 3-axis acceleration data from sensors worn on the wrist. Following Yuan et al. [22], we divided the signal into fixed-length sliding windows and treated them as independent inputs. All data were linearly resampled to a frequency of 30Hz using a 10-second sliding window. The dataset used is summarized in Table 1.

The pre-training data consists of unlabeled data from ssl-capture-24 [3]. This dataset includes a total of 909,535 sliding windows. The downstream tasks include ADL [1], Opportunity [13], PAMAP2 [12], REALWORLD [17], and WISDM [19], as shown in Table 1. We evaluated our methods using these five datasets.

The classification results for human activity recognition, a downstream task, are presented in Table 2. For comparison, we include results from SENvT-u4 and SENvT-u7 by Okita et al. [8]. Note the similarities between Transformer settings and our approach: we use unlabeled training data in the initial step (similar to LLM, PLM, or pre-trained model construction) and labeled test data with the (pre-)trained model in the second step (downstream task).

For RepcondFormer, multitask training did not improve performance across all datasets in transfer learning, whereas fine-tuning improved the F1 score across all datasets. Specifically, for the Opportunity dataset, accuracy improved by 3.41% and the F1 score by 1.19% compared to SENvT-U7. For RepcondClassifier, improvements for the PAMAP2 dataset were 2.86% in accuracy and 3.23% in the F1 score compared to SENvT-U7.

4.2 Evaluation of RepcondFormer

Fine-tuning improves performance across four datasets, while transfer learning did not show improvements on any dataset. Figure 4 illustrates the low-dimensional visualization of signal classes using t-SNE, with the same signal class represented by the same color. The left figure shows the raw data, consisting of 900 dimensions obtained by concatenating a 300×3 sliding window series by channel. The central figure shows the output from RepcondFormer, with parameters fixed after training the diffusion transformers. The right figure displays the representation from the representation-conditional diffusion transformer after fine-tuning with the initialized RepcondFormer.

The central figure represents the output used for transfer learning, trained through a linear layer with these representations, which

[3]Formatted by Yuan et al. [22]: 'https://zenodo.org/records/6574265' zip file 'data/ssl/ssl_capture_24/data/train/P*.npy'.

Table 2: The figure displays the results for the downstream task of human activity recognition. We compared our methods against baseline classifiers, including a random forest and a transformer trained from scratch. SENvT-X represents the results obtained using a vision transformer-based model. The results for RepcondFormer and RepcondClassifier are provided by our proposed methods.

Method	ADL		Opportunity		PAMAP2		REALWORLD		WISDM	
	acc	f1	acc	f1	acc	f1	acc	f1	acc	f1
Random Forest[1]	.8346	.6830	.6814	.4260	.6602	.5467	.6618	.5712	.4630	.4224
Transformer scratch[2]	.8503	.8230	.6276	.6209	.7682	.7594	.8554	.8663	.8219	.8194
transfer learning										
SENvT-u4[8]	.8394	.7936	.7103	.6826	.7157	.6912	.7470	.7512	.6667	.6618
SENvT-u7[8]	.8472	.7849	.7150	.6909	.7098	.6869	.7628	.7692	.6819	.6766
RepcondFormer	.7858	.6205	.6033	.5024	.6753	.6440	.7022	.7135	.6108	.5945
fine-tuning										
SENvT-u4[8]	.9078	.8780	.7489	.7411	.8488	.8454	.9015	.9124	.8693	.8679
SENvT-u7[8]	.8992	.8684	.7429	.7448	.8477	.8445	.9058	.9163	.8875	.8867
RepcondFormer	.9055	.8826	.7770	.7567	.8554	.8505	.9089	.9188	.9016	.9008
RepcondClassifier	.8818	.8515	.7693	.7526	.8763	.8768	.8959	.8574	.8076	.8075

*1 The input data had 900 dimensions, with x, y, and z axes concatenated in series, a depth of 5, and 10 trees. Classification results were obtained using this configuration.

*2 Classification results were obtained using a model with a depth of 6, an embedding dimension of 128, and 4 attention heads, with a learning rate of 0.001 and 50 epochs.

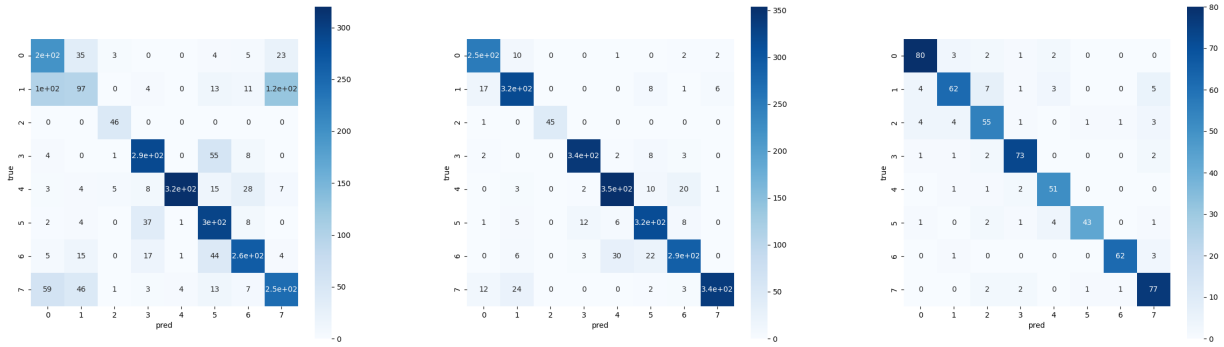

Figure 3: The figure shows the confusion matrices for the RepcondFormer classifier on the REALWORLD dataset and the RepcondClassifier on the PAMAP2 dataset. The upper matrix represents results from RepcondFormer using transfer learning, the middle matrix shows results from RepcondFormer using fine-tuning, and the lower matrix displays results from RepcondClassifier.

yielded suboptimal results. In contrast, fine-tuning improved results with the representations shown in the central figure. This suggests that the data representation optimized by the representation-conditional diffusion model may not offer significant advantages when used directly in classification tasks.

4.3 Evaluation of RepcondClassifier

Improvements were observed for PAMAP2, while results for WISDM were less satisfactory. This discrepancy may be due to the number of classes: PAMAP2 has 8 classes, whereas WISDM has 18. Similar signals may cluster within the same class label. As the number of classes increases, signals within the same class may become more mixed, complicating classification. RepcondClassifier may misclassify signals if they do not closely resemble the class label, even if they are from the same class.

Additionally, when using RepcondClassifier, the source noise and signal are separated by the denoising network of DiT. If the number of samples is small, the expected results may not be reliable. While it is ideal to use as many samples as possible, this experiment used a sample size of 64.

4.4 Evaluation of Signal Generation

Following the approach of Smith et al. [15], who evaluated image generation quality, we adapted the model to learn from signal

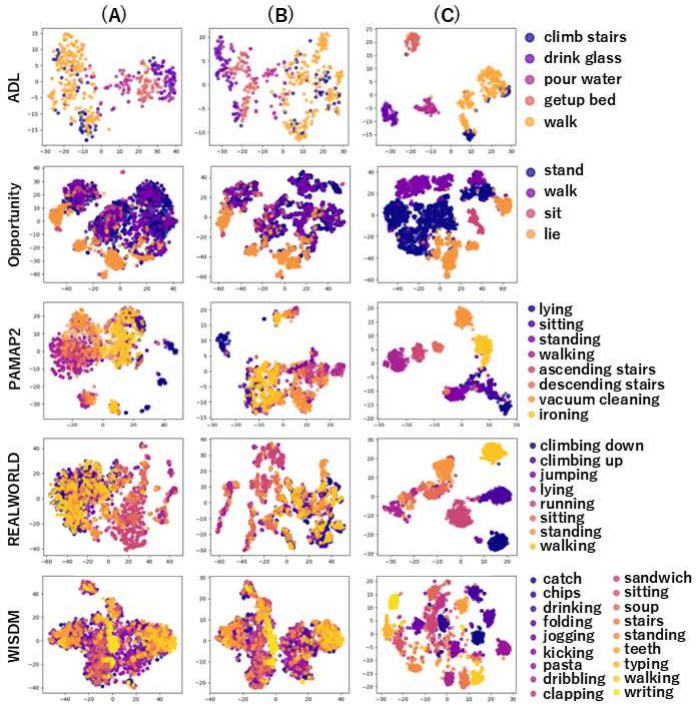

Figure 4: The figure displays signal features with class labels mapped into low dimensions using t-SNE. All five datasets are plotted from top to bottom. The left column (A) shows the raw data. The second column (B) depicts the representation encoded by RepcondFormer after training the diffusion models. The right column (C) shows the representation encoded by RepcondFormer after fine-tuning.

data instead of image data in a supervised learning context. While Inception-V3 [16] is commonly used in image generation, we employed a modified version of this model for one-dimensional input. We used FID, IS, precision, and recall metrics, calculated based on features obtained from this model.

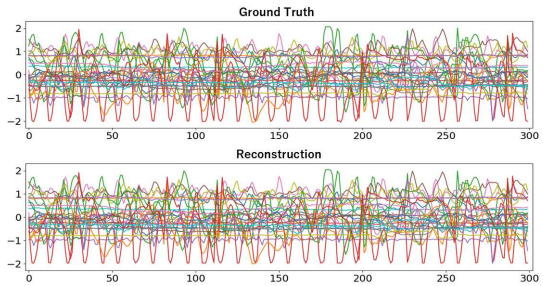

Figure 5: The reconstruction of signals using a diffusion transformer with representation conditioning is illustrated. The upper figure shows the original data, which serves as the ground truth. The lower figure displays the signal generated based on the original data.

Table 3: The table presents the evaluation results for generative models. We assessed both the unconditional diffusion transformer and the representation-conditioned diffusion transformer.

Conditions	Training steps	FID↓	IS↑	Precision↑	Recall↑
None	500k	70.75	2.644	0.249	0.249
Representation	500k	0.339	2.930	0.988	0.989

Table 3 presents the results of reconstructing 50,000 signal data samples against their ground-truth data. It compares an unconditionally trained model with a model trained using representation-conditional methods.

Figure 5 shows the reconstruction of signals by representation-conditional diffusion transformers. The top section displays the original data, while the bottom section shows the signal generated through reconstruction. The differences are minimal, and the generated signals closely resemble the original data.

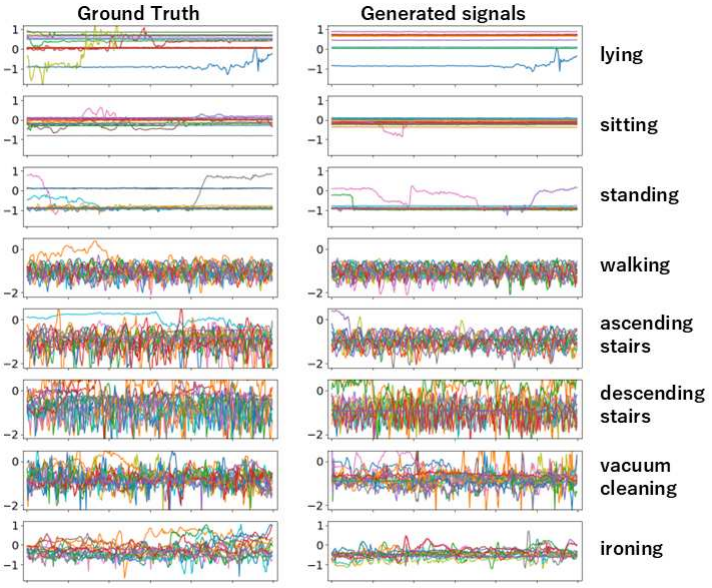

Figure 6: The figure shows signals generated for each class in the PAMAP2 dataset. It includes eight activity classes. The left panel displays signals randomly extracted from the eight ground-truth datasets, while the right panel shows signals generated by conditioning on each class label.

Figure 6 illustrates signals generated for specific class labels from the perspective of signal generation. When using Repcond-Classifier, we fine-tune the diffusion transformer with class labels. Subsequently, signals are generated conditioned on these class labels. The figure presents results using the PAMAP2 dataset, which consists of 8 activity labels: *lying*, *sitting*, *standing*, *walking*, *ascending stairs*, *descending stairs*, *vacuum cleaning*, and *ironing*. The activity labels are plotted from top to bottom. The left column shows randomly extracted signals from the original data, while the right column displays signals generated conditioned on the class labels.

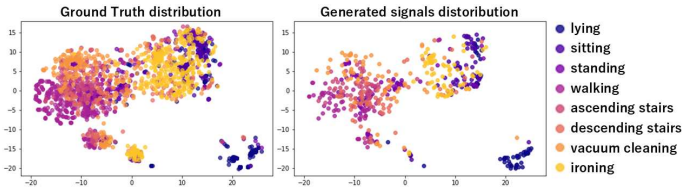

Figure 7: The figure also presents a two-dimensional plot of signals generated for each class, using the PAMAP2 dataset. The left plot depicts the distribution of the original data (ground truth), while the right plot illustrates the distribution of signals generated for each class label.

The second row from the bottom in Figure 6 highlights that the *vacuum cleaning* sensor signals were not well-reconstructed, as indicated by the lack of dispersion in the orange color compared to the distribution of the original data.

Figure 7 provides a similar comparison using the PAMAP2 data from a classification perspective. The right figure shows signals generated conditioned on the class labels, while the left figure displays the original data.

In general, diffusion models show promising performance in terms of reconstruction and FID. However, reconsidering the evaluation results of the two classifiers suggests that the tuned class-conditional diffusion transformers may not have performed optimally. Specifically, these class-conditional diffusion transformers struggled with generating accurate signals for the *vacuum cleaning* label in the PAMAP2 dataset.

5 Conclusion

This paper introduces the representation-conditional diffusion transformer and two types of classifiers. Both classifiers performed exceptionally well in the fine-tuning setting. RepcondFormer demonstrated superior performance in fine-tuning, although its performance slightly decreased in transfer learning.

Several future directions are proposed. First, the sample size for RepcondClassifier was set to 64. A small sample size may lead to unreliable results due to sampling uncertainties from the probability density function. Increasing the sample size could mitigate this issue and provide better insights into optimal values. Second, while representation-conditional diffusion transformers achieved high performance in terms of FID and reconstruction, class-conditional diffusion transformers did not. This indicates a need to refine tuning methods by fixing certain parameters rather than relying on simple tuning. Third, the original DiT operates on images within a latent space with compressed images. Our study did not explore the effects of compression in the latent space. Integrating sensor data into a latent space mechanism (e.g., TimeVQVAE [6]) could potentially improve results and enhance the DiT's performance.

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